# D3.2 Inheritance

*Notes written for this site. Slides, videos, worksheets and self-check questions: whatthecell.org/subtopics/sl-d3-2.html*

*Written for this site from the course textbook and class slides, with page references for each statement. The chapter is on pages 725–746 of the course textbook. Image credits are on the last slide of the slides for this subtopic.*

## THE BASICS

### D3.2.1 Gametes and zygotes

p. 725

Parents make **haploid gametes**, with one copy of each gene, and these fuse to form a **diploid zygote**, with two copies of each autosomal gene. This pattern is common to all eukaryotes with a sexual life cycle.

### D3.2.2 Genetic crosses in flowering plants

pp. 726–730

Pollen contains male gametes and female gametes are in the ovary, so a cross needs **pollination**. Peas make both on one plant, so they can self-fertilize. The parents are the **P generation**, their offspring are **F1**, and the offspring of F1 are **F2**. Crosses are widely used to breed new varieties.

### D3.2.3 Genotype

p. 730

A **gene** influences a characteristic, and an **allele** is a version of a gene. The **genotype** is the combination of alleles inherited. It is **homozygous** if the two alleles are identical, and **heterozygous** if they differ.

### D3.2.4 Phenotype

p. 731

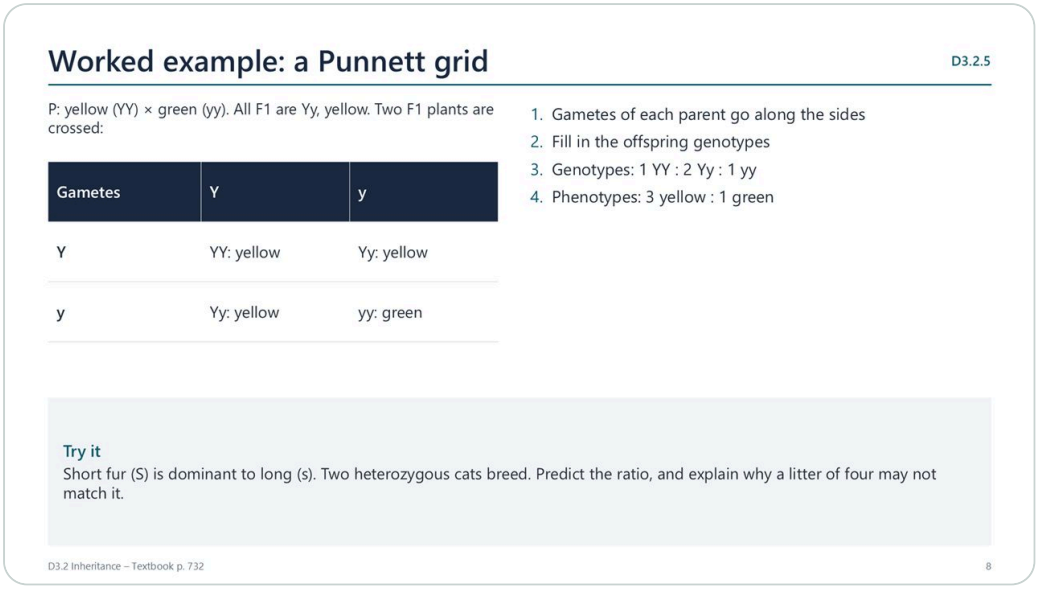
The **phenotype** is the observable traits, from genotype and environment. ABO blood group depends on genotype only, a surgical scar on environment only, and height on both.

### D3.2.5 Dominant and recessive alleles

p. 732

A **dominant** allele has the same effect whether homozygous or heterozygous. A **recessive** allele has an effect only when homozygous. YY and Yy look alike because the dominant allele usually codes for a working protein, and one copy gives enough.

Crossing two heterozygotes gives a 3 : 1 phenotype ratio.



**Worked example: a Punnett grid** D3.2.5

P: yellow (YY) × green (yy). All F1 are Yy, yellow. Two F1 plants are crossed:

| Gametes | Y | y |
|---|---|---|
| Y | YY: yellow | Yy: yellow |
| y | Yy: yellow | yy: green |

1. Gametes of each parent go along the sides
2. Fill in the offspring genotypes
3. Genotypes: 1 YY : 2 Yy : 1 yy
4. Phenotypes: 3 yellow : 1 green

**Try it**
Short fur (S) is dominant to long (s). Two heterozygous cats breed. Predict the ratio, and explain why a litter of four may not match it.

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## EXPRESSION AND DISEASE

### D3.2.6 Phenotypic plasticity

p. 732

Organisms can develop traits suited to their environment by varying **gene expression**. The genotype does not change, and the effect may be reversible, as when skin darkens in sunlight and then fades.

### D3.2.7 Phenylketonuria

p. 733

**PKU** is a recessive condition, due to a mutation in an autosomal gene for the enzyme that converts phenylalanine to tyrosine. Without it, phenylalanine builds up and can damage the developing brain. Babies are screened after birth, and a low-phenylalanine diet prevents most harm.

## BEYOND TWO ALLELES

### D3.2.8 SNPs and multiple alleles

p. 734

A **gene pool** is all the alleles in a population. SNPs create new alleles, so any number of alleles of a gene can exist in a gene pool. An individual still inherits only two.

### D3.2.9 ABO blood groups

pp. 734–735

| GROUP | GENOTYPES |
|---|---|
| A | $I^AI^A$ or $I^Ai$ |
| B | $I^BI^B$ or $I^Bi$ |
| AB | $I^AI^B$ |
| O | ii |

### D3.2.10 Incomplete dominance and codominance

pp. 736–737

In **incomplete dominance**, heterozygotes have an intermediate phenotype: red and white *Mirabilis jalapa* give pink. In **codominance**, they have a dual phenotype: $I^AI^B$ gives group AB. Crossing two heterozygotes gives a 1 : 2 : 1 ratio.

## SEX AND PEDIGREES

### D3.2.11 Sex determination

pp. 737–738

All eggs carry an X, and half of sperm carry a Y. So the sex chromosome in the sperm determines whether a zygote develops certain male-typical or female-typical characteristics. The X chromosome carries far more genes than the Y.

### D3.2.12 Haemophilia

pp. 739–740

The recessive allele $X^h$ is on the X chromosome. Males have only one X, so one copy causes the disease. A carrier mother ($X^HX^h$) and an unaffected father ($X^HY$) have a 1 in 2 chance that a son is affected.

### D3.2.13 Pedigree charts

pp. 741–743

- **Autosomal recessive:** unaffected parents can have an affected child, of either sex.
- **Autosomal dominant:** every affected person has an affected parent.
- **X-linked recessive:** mostly males are affected, never passed from father to son.

Close relatives are more likely to carry the same recessive allele. This is the genetic basis for rules in many societies against marriage between close relatives.

## CONTINUOUS TRAITS

### D3.2.14 Continuous variation

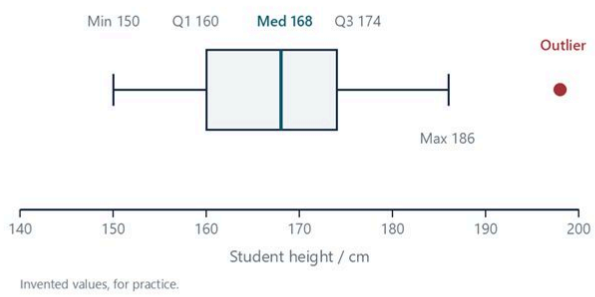
pp. 743–745

ABO blood group is a **discrete** variable, and skin colour is a **continuous** one. **Polygenic** traits are affected by many genes, each with a small effect, often with the environment as well. The centre of a distribution can be described by its mean, median and mode.

## D3.2.15 Box-and-whisker plots

### Box-and-whisker plots

D3.2.15




Invented values, for practice.

A box plot shows the **median**, **quartiles**, **range** and any **outliers**.

**IQR** = Q3 − Q1. An outlier is more than 1.5 × IQR beyond a quartile.

**Worked example**

IQR = 174 − 160 = 14; 1.5 × 14 = 21. Upper limit 174 + 21 = 195, so 198 cm is an outlier, and the whisker stops at 186 cm.

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A plot shows outliers, minimum, first quartile, median, third quartile and maximum. A value is an outlier if it is more than 1.5 × IQR above Q3 or below Q1.

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